METRIC

FF-B-171/13

30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING, ANSI/AFBMA DIMENSION SERIES 22.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

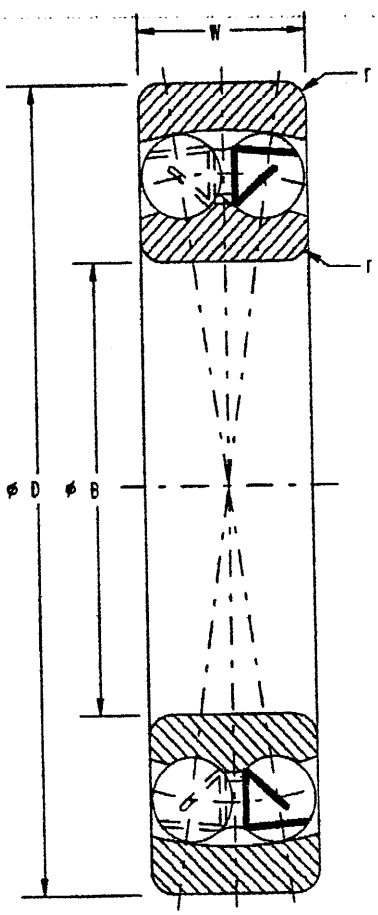


FIGURE 1.
DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING WITH CYLINDRICAL BORE (CODE C)

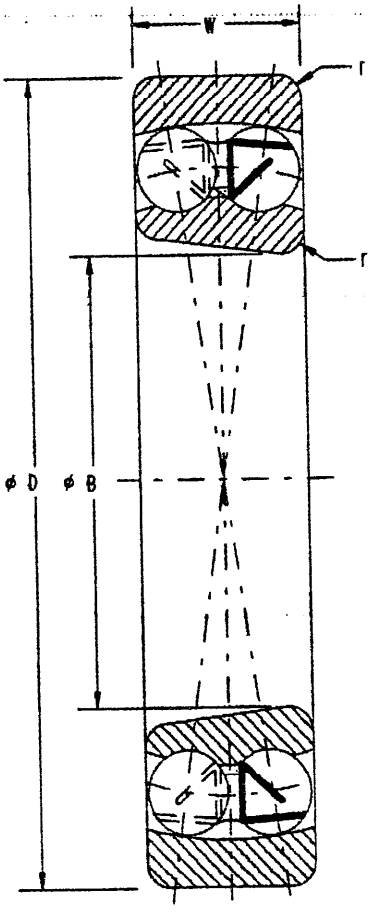


FIGURE 2.
DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING WITH TAPERED BORE (CODE T)

BEARING CONFIGURATIONS

FSC 3110

AMSC/NA

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Design. All self-aligning bearings shall be open. Bores shall be cylindrical or tapered (taper 1:12) and shall be specified in the part number by the codes in table III.

4. Misalignment. Maximum permissible misalignment shall be 4.0 degrees.

5. Radial Internal clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

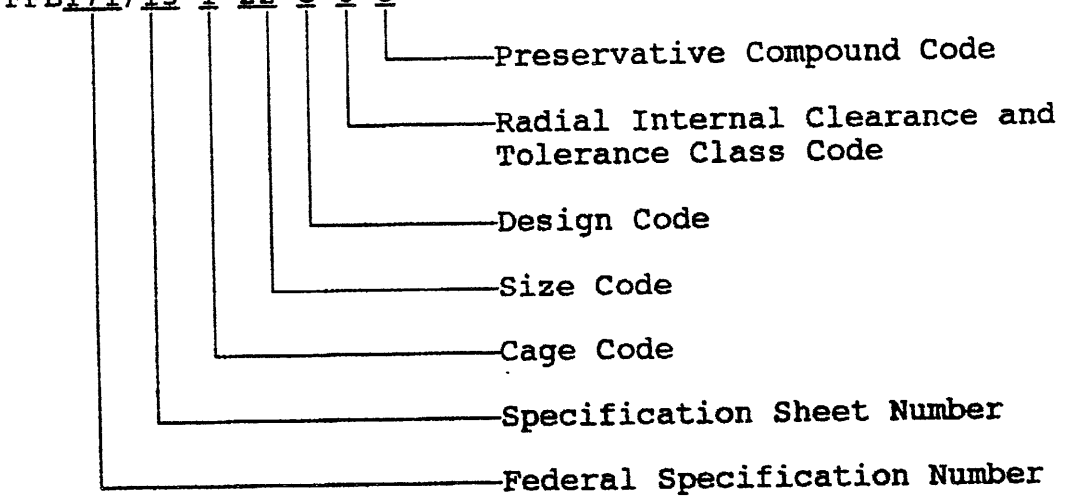
6. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

7. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code "C" in the part number.

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8. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, bore design, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE: FFB171/13 T 22 C 6 C



FFB171/13T22C6C indicates - Bearing, cage-machined nonmetallic (phenolic); bore 110mm; outside diameter 200mm; width 53mm; open (no shields or seals), cylindrical bore; radial internal clearance symbol 0 (normal); tolerance class ABEC-5; preservative compound in accordance with MIL-P-197.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	30	14	0.6	15	1100
01	12	32	14	0.6	17	1300
02	15	35	14	0.6	20	1315
03	17	40	16	0.6	22	1685
04	20	47	18	1.0	26	2145
05	25	52	18	1.0	31	2151
06	30	62	20	1.0	36	2753
07	35	72	23	1.1	42	3760
08	40	80	23	1.1	47	3975
09	45	85	23	1.1	52	3980
10	50	90	23	1.1	57	4120
11	55	100	25	1.5	63.5	4580
12	60	110	28	1.5	68.5	5870
13	65	120	31	1.5	73.5	7520
14	70	125	31	1.5	78.5	7610
15	75	130	31	1.5	83.5	7685
16	80	140	33	2.0	90	8560
17	85	150	36	2.0	95	10075
18	90	160	40	2.0	100	12105
19	95	170	43	2.1	107	14375
20	100	180	46	2.1	112	16570
21	105	190	50	2.1	117	18870
22	110	200	53	2.1	122	20715
24	120	215	58	2.1	132	23130
26	130	230	64	3.0	144	26870
28	140	250	68	3.0	154	28640
30	150	270	73	3.0	164	33485

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

TABLE III. Design Codes.

Code	Design
C	Open (no shields or seals), Cylindrical bore - fig.1
T	Open (no shields or seals), Tapered bore - fig.2

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force 11

Review
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

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